Bandana Dhakal

Chitwan, Bagmati, Nepal | bandanadhakal156@gmail.com |

LinkedIn | GitHub | Portfolio

Professional Summary

BSC Physics graduate from Shaheed Smriti Multiple Campus, Institute of Science and Technology (IOST), Tribhuvan University, Nepal, with a strong academic record and research experience in household energy consumption, fuel stacking, and sustainable energy transitions. Ranked first in the Physics programme throughout all four academic years and received a departmental merit scholarship. Experienced in quantitative data analysis, statistical modelling, and scientific computing using Python and MATLAB.

Experience

Lead Author and Thesis Researcher | March 2026 to August 2026

Shaheed Smriti Multiple Campus, IOST, Tribhuvan University - Chitwan, Nepal

- Led undergraduate research investigating household energy consumption, fuel stacking, and electricity adoption across 201 households in two semi-urban settlements in Chitwan, Nepal.
- Conducted statistical analysis using Python (pandas, NumPy, SciPy, Matplotlib) to derive insights from data, including descriptive statistics, logistic regression, and sensitivity analysis.
- Developed computational models to estimate useful cooking energy consumption, determining average consumption of 246 MJ per household per month from LPG.
- Applied scientific computing and quantitative research methods to interpret findings and effectively communicate research outcomes.

Education

Bachelor of Science (BSc) in Physics | Completed October 2026

Shaheed Smriti Multiple Campus, Institute of Science and Technology (IOST), Tribhuvan University - Chitwan, Nepal

- Ranked first in the Physics department throughout all four years of the Bachelor of Science programme (2022-2026).
- Completed a final-year research thesis titled “Fuel Stacking and Electricity Use in Two Semi-Urban Settlements in Nepal,” achieving a score of 92, the highest among the graduating cohort.
- Conducted quantitative research involving 201 households, applying Python programming, statistical analysis, logistic regression, and numerical modelling.
- Developed expertise in classical mechanics, quantum physics, electromagnetism, thermodynamics, statistical physics, and mathematical physics.
- Awarded a full departmental Academic Excellence Scholarship throughout the four-year undergraduate programme in recognition of exceptional academic performance.

Higher Secondary School: +2 Science | 2021

Nepal Higher Secondary School - Chitwan, Nepal

- Graduated with a GPA of 3.43/4.00 in the Higher Secondary Education (+2 Science) programme with majors in maths and Physics.
- Achieved first rank among students in the +2 Science programme, demonstrating outstanding academic performance.
- Awarded a full academic scholarship in recognition of exceptional academic achievement and first-place ranking.
- Developed a strong academic foundation in Physics, Mathematics, Chemistry, and scientific problem-solving.
- [Award] Awarded a full academic scholarship in recognition of exceptional academic achievement and first-place ranking.

Secondary Education Examination (SEE) | June 2019

Ekata Shishu Niketan - Chitwan, Nepal

- Completed the Secondary Education Examination (SEE) with a GPA of 3.50 out of 4.00.
- Developed a strong academic foundation in Mathematics, Science, English, and other core subjects.
- Demonstrated academic dedication, analytical thinking, and problem-solving abilities throughout secondary education.

Research Projects

Air Quality Analysis and PM2.5 Prediction in Ratnanagar, Nepal (2025)

Independent computational research project | Python, Google Colab, Scikit-learn

• Analyzed 8,760 hourly modelled Open-Meteo air-quality observations (2025), including PM2.5, PM10, nitrogen dioxide, ozone, and US AQI; examined seasonal trends and pollutant correlations.

• Built and compared one-hour-ahead PM2.5 forecasts using persistence baseline, Linear Regression, and Random Forest with a chronological 80:20 train/test split.

• Random Forest achieved MAE 3.032 μg/m^3, RMSE 4.878 μg/m^3, and R^2 0.963 on held-out modelled data; results are not independently validated against local monitoring stations.

• Project repository: https://github.com/bandanadhakal156-cpu/Ratnanagar-Air-Quality-2025

Skills

Python • GitHub • LaTeX • Microsoft Excel and MS Word • PowerPoint • Computational physics • Scientific computing • Numerical analysis • Mathematical modeling • Data science • JAVA • HTML • C • C++

References

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Languages

- English
- Nepali
- Hindi